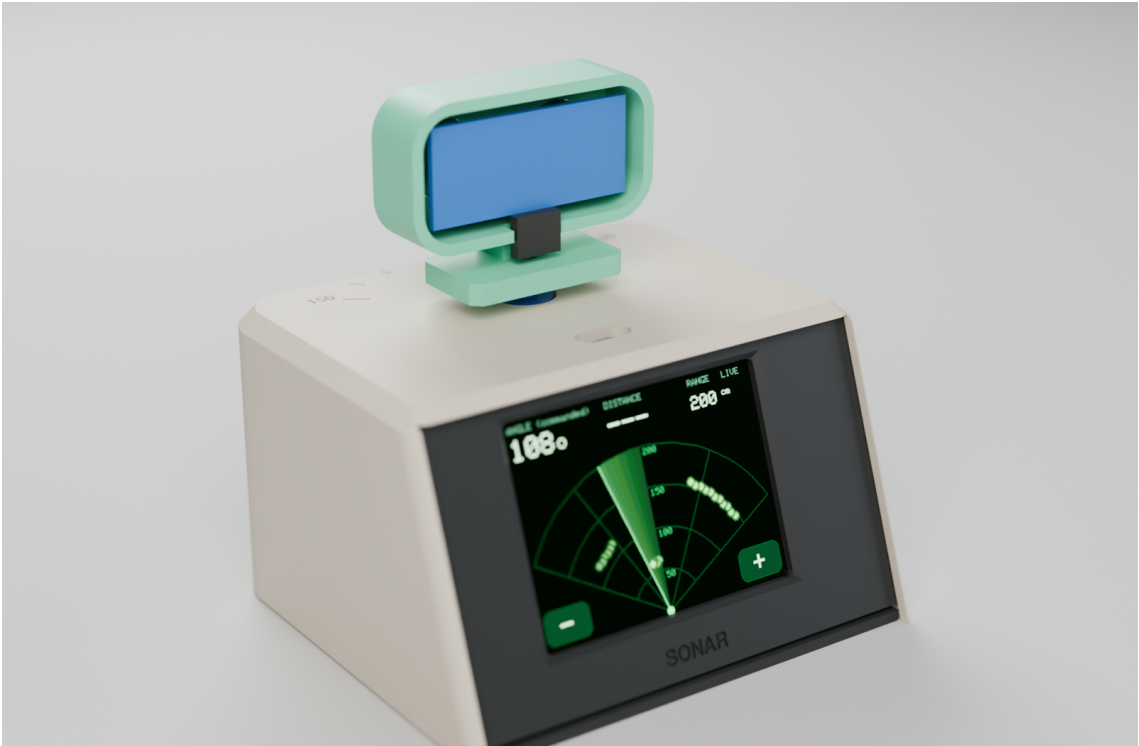


Ultrasonic Radar Scanner

Build packet — Radar V6 “Mini” · repository release 6.0.0 · generated 2026-10-04



Honest scope. An educational ultrasonic (sonar) instrument with a radar-style display — not radio-frequency radar and not a safety, obstacle-avoidance or people-detection device. It sweeps about 120°; the angle shown is the commanded servo position. At the time this packet was generated, no unit had been printed, wired or powered — your build is part of the validation.

Unit / serial	
Builder	
Start date	
Printer + filament	
Firmware (version / SHA-256)	
Notes	

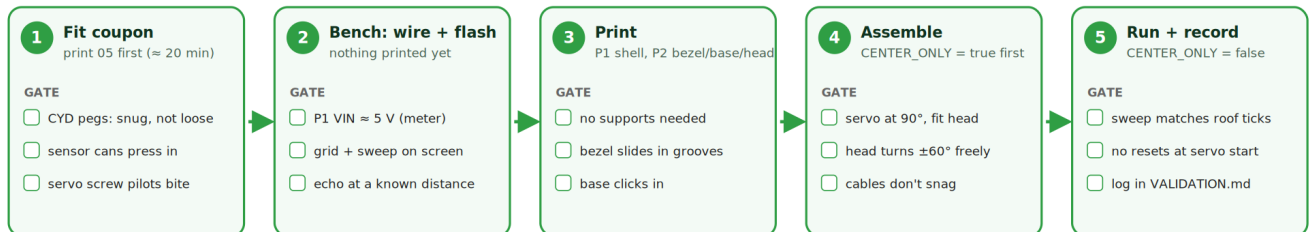
1 · Safety rules (read before wiring)

- ☐ Wire only with USB **unplugged**. Check every wire against the list before plugging in.
- ☐ Use a **3.3 V-capable** sensor (RCWL-1601 / HC-SR04P). A 5 V-only HC-SR04 needs the 2.2 kΩ / 3.3 kΩ divider on ECHO (docs/WIRING.md, fallback).
- ☐ Measure **P1 VIN** $\approx$ **5 V** with a multimeter before connecting the servo to it.
- ☐ Leave P1 TX/RX and P3 IO21/IO22 unconnected.
- ☐ Power from a charger or power bank rated $\geq$ 1 A. Unplug if anything gets hot or smells.
- ☐ Keep fingers and hair clear of the turning head; it is a toy-grade servo, not a guarded mechanism.

Build stages — each has a gate; record results before moving on

Build stages — do not skip a gate

Each stage has a pass/fail gate. Record the result in docs/VALIDATION.md before moving right. A failed gate means: adjust the named VERIFY variable, re-render, repeat.



2 · Parts checklist

The whole kit — 5 parts to buy, 1 optional

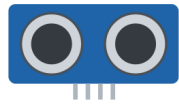
Plus a USB cable, a ≥ 1 A charger and ≈ 120 g PLA. Prices are rough single-unit estimates (US\$), not quotes. Generated from docs/BOM.md.



1× CYD board

ESP32 + 2.8" touch screen in one

E1 · $\approx$ \$12-20



1× Sensor

RCWL-1601 or HC-SR04P (3.3 V)

E2 · $\approx$ \$2-4



1× SG90 servo

positional 180°, horn + 2 screws

E3 · $\approx$ \$2-3



3× Cables

JST 1.25 mm 4-pin → Dupont female

E4 · $\approx$ \$1-2



3× Jumper pins

male-male, join the servo plug

E5 · $<$ \$1



1× Capacitor

470-1000 μ F, optional

E6 · $<$ \$1

✓	ID	Qty	Part	Check before you buy
☐	E1	1	ESP32-2432S028R "Cheap Yellow Display" (CYD) — ESP32 + 2.8" 320×240 touch screen	Exact name ESP32-2432S028R. Both the ILI9341 version and the two-USB "CYD2USB" (ST7789) version work — the firmware detects the panel.
☐	E2	1	Ultrasonic sensor, 3.3 V capable: RCWL-1601 or HC-SR04P	Listing must say it works from 3 V-5.5 V. A plain 5 V-only HC-SR04 needs the fallback divider.
☐	E3	1	SG90 micro servo, positional (180°), with horns and its 2 screws	Not "360°" or "continuous rotation". The single-arm horn must come in the bag.
☐	E4	3	JST 1.25 mm 4-pin → Dupont female cable	Often one is in the CYD box — you need three (CN1, P3, P1).
☐	E5	3	Male-male Dupont jumper pins (or 3 short male-male jumpers)	Join the servo plug to the cable.
☐	E6	1	470-1000 μF, ≥ 6.3 V electrolytic capacitor	Only if the board resets when the servo moves.
☐	F1	2	self-tapping screws that come with the SG90	servo flange → the two bosses under the roof
☐	F2	1	small screw that comes with the SG90 horn (optional)	horn → head foot, only if the press fit is loose

3 · Printed parts

Files: **cad/3mf/** (Bambu A1 mini plates), **cad/stl/** (already in print orientation), **cad/step/** (editable). Print the fit coupon first and adjust any VERIFY value in cad/build_radar.py before the long prints. PLA, 0.20 mm layers, 3 walls, 15–20 % gyroid, **no supports**.

✓	Part	Role	Plate	Print size (mm)	PLA ≤ g	Orientation
☐	01_shell	required	P1	104.0 x 94.9 x 62.0	64.2	roof on the plate; 45-degree chamfers, no supports
☐	02_bezel	required	P2	100.7 x 59.6 x 8.4	10.8	viewer face on the plate; pegs up
☐	03_base	required	P2	99.5 x 90.1 x 7.4	28.3	flat; rim up
☐	04_head	required	P2	56.0 x 39.5 x 20.0	13.0	face on the plate; neck and foot lie flat
☐	05_fit_coupon	print first - test piece	S1	92.0 x 42.0 x 9.0	12.0	flat; print this first

Plate S1_fit_coupon - Bambu Lab A1 mini (180 x 180 mm)

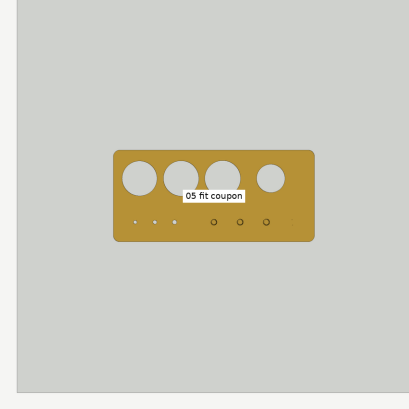
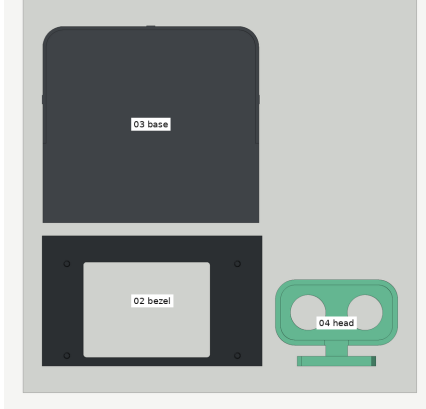


Plate P1 - Bambu Lab A1 mini (180 x 180 mm)



Plate P2 - Bambu Lab A1 mini (180 x 180 mm)



Fit coupon (stage 1)

✓	Coupon test	Pass when	Size chosen
☐	CYD pegs 2.8 / 2.9 / 3.0 mm	board presses on and stays; no cracking	
☐	Sensor can holes 16.0 / 16.2 / 16.4 mm	cans press in and hold, PCB not stressed	
☐	Servo screw pilots 1.6 / 1.8 / 2.0 mm	the SG90 screw bites and holds without splitting	

4 · Wiring — point to point

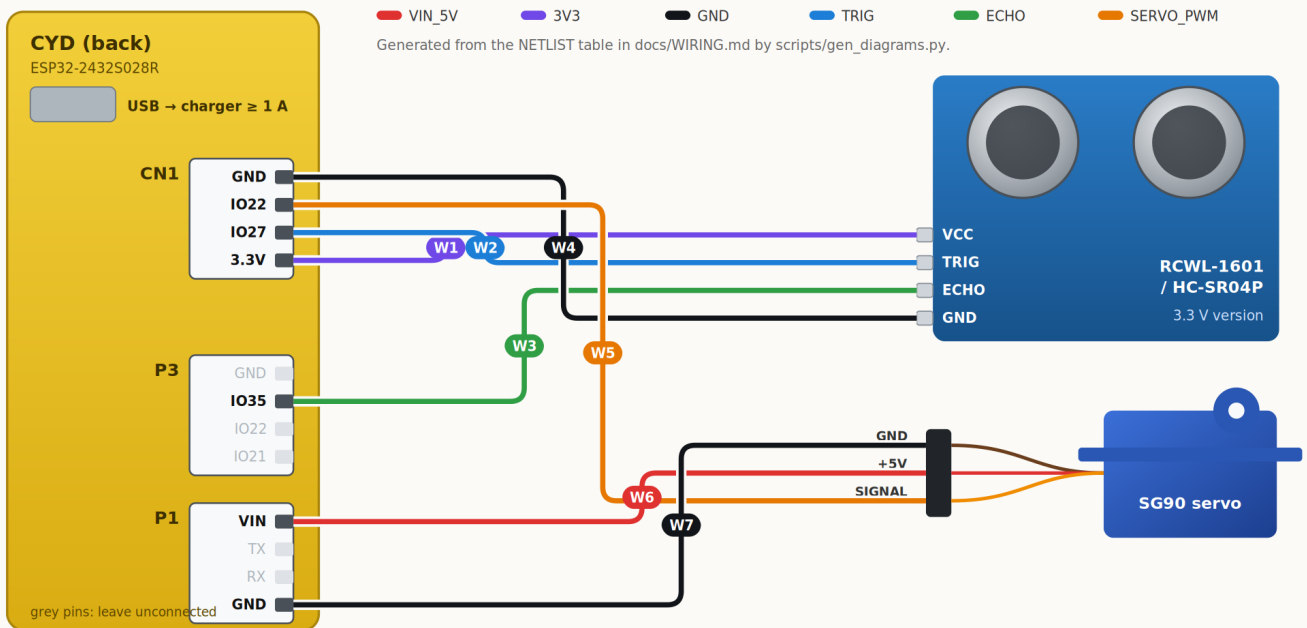
Cut, label and fit each wire, then tick it. Follow the **labels** on the modules, not wire colours. Seven wires; USB unplugged. Source of truth: docs/WIRING.md.

✓	Wire	From	To	Net	Notes
☐	W1	CN1.3V3	SENSOR.VCC	3V3	3.3 V from the CYD regulator; RCWL-1601 / HC-SR04P runs at 3.3 V
☐	W2	CN1.IO27	SENSOR.TRIG	TRIG	10 µs trigger pulse
☐	W3	P3.IO35	SENSOR.ECHO	ECHO	input-only pin; 3.3 V echo, no divider
☐	W4	CN1.GND	SENSOR.GND	GND	common ground
☐	W5	CN1.IO22	SERVO.SIGNAL	SERVO_PWM	50 Hz pulses, 1000–2000 µs; orange (or yellow/white) lead
☐	W6	P1.VIN	SERVO.+5V	VIN_5V	USB 5 V on the board side. VERIFY on your board; red lead (middle)
☐	W7	P1.GND	SERVO.GND	GND	same copper as CN1.GND; brown (or black) lead

Not connected on purpose: P1 TX/RX (USB serial), P3 GND/IO22/IO21, and the CYD's SD card and speaker. The servo plug and the cables often both end in female sockets — join them with male-male pins.

7 wires. No breadboard, no resistors, no soldering to the board.

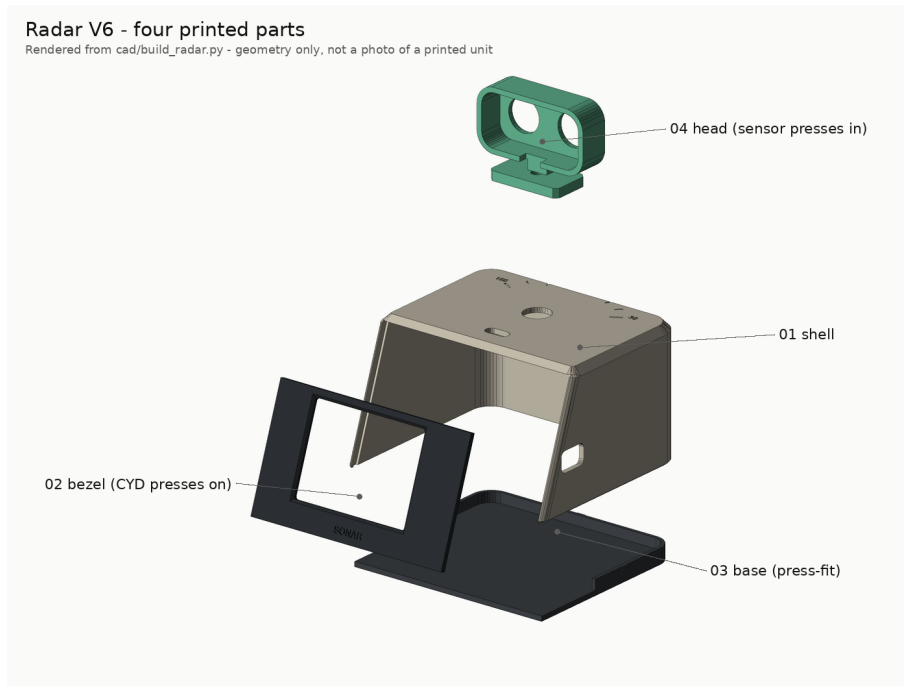
Back of the Cheap Yellow Display (CYD). Illustration, not a photo — pin order on your connectors may differ: follow the silkscreen.



5 · Firmware

- ☐ Connect the CYD to the computer with a data-capable USB cable (sensor and servo may stay connected).
- ☐ EITHER open the web page → **Flash firmware** (Chrome/Edge) and install the release build, OR compile: `arduino-cli compile --upload --profile esp32-core3 -p <port> firmware/Radar_V6`.
- ☐ For assembly, first flash with **CENTER_ONLY = true** (servo holds 90°), then set it back to false.
- ☐ Splash shows “EDU SONAR ... NOT a safety device”, then the green fan and the sweep.

6 · Assembly gates



Stage 2 — bench (nothing printed yet)

- ☐ P1 VIN $\approx$ 5 V measured with USB in
- ☐ Grid, sweep and readings appear; touch -/+ changes range
- ☐ Echo at 30 / 100 / 150 cm recorded
- ☐ No resets when the servo starts (else add the capacitor)

Stage 3 — print

- ☐ P1 shell and P2 bezel/base/head printed without supports
- ☐ Bezel slides into the shell grooves without forcing

Stage 4 — assemble

- ☐ CYD pressed onto the bezel pegs, screen through the window
- ☐ Servo screwed under the roof with its own 2 screws
- ☐ CENTER_ONLY: horn pressed into the head foot, head facing the scan side
- ☐ Sensor cans pressed into the head; cable through the slot with slack
- ☐ Bezel slid in, base pressed in; USB socket reachable

Stage 5 — run

- ☐ CENTER_ONLY = false; head sweeps 30°-150° without touching anything
- ☐ Screen direction matches the head (else REVERSE_SERVO)
- ☐ Roof ticks match the screen angle (CALIBRATE_SERVO)

7 · Physical test sheet

Copy results into docs/VALIDATION.md (or open a pull request) — that is how the design becomes validated.

Test	What	Pass criterion	Result	Date / initials
T-PRN-1	Print the fit coupon; try the CYD pegs, sensor-can holes and servo-screw pilots	chosen sizes recorded; PEG_D, CAN_HOLE_D, PILOT_D updated if needed		
T-PRN-2	Print P1 (shell, roof-down)	no supports, chamfers and grooves clean		
T-PRN-3	Print P2 (bezel, base, head)	no supports; pegs not snapped		
T-FIT-1	CYD pressed onto the bezel pegs	firm; screen centred in the window (else set CYD_VIEW_DX)		
T-FIT-2	Bezel slides into the shell grooves; base presses in; USB plug reaches the board	nothing forced; base stays in when lifted		
T-FIT-3	Servo screwed under the roof with its own screws; horn pressed into the head foot	head firm on the spline, no wobble		
T-FIT-4	Sensor cans pressed into the head	cans hold; PCB not bent		
T-PWR-1	P1 VIN voltage with USB in: idle, and while the servo moves	≈ 4.5–5.2 V; record the dip		
T-PWR-2	One-hour run from a ≥ 1 A charger	no resets, flicker or brown-outs (else add the capacitor and re-test)		
T-PWR-3	Run from a laptop USB port	record whether it works; note resets		
T-PWR-4	Current draw with a USB meter: idle and sweeping	record mA		
T-FW-1	Flash; splash, then grid and sweep	as described in ASSEMBLY		
T-FW-2	Panel detection and colours (# board ... panel ... line)	correct picture; else record the PANEL / INVERT_OVERRIDE / SWAP_RED_BLUE that work		
T-FW-3	CENTER_ONLY head fit; REVERSE_SERVO direction	head faces straight out at 90°; screen matches head		
T-FW-4	Measured timing from the serial "# measured" line	record mean step and one-way sweep (estimate ≈ 115 ms, ≈ 4.6 s)		
T-FW-5	Distance at 30 / 100 / 150 cm on the 90° line (flat target)	record reading vs tape measure		
T-FW-6	Touch: – / + change the range; tap the fan to pause	each tap acts once; no false taps		
T-FW-7	RGB LED turns red under 30 cm	as described		
T-CAL-1	CALIBRATE_SERVO against the roof ticks without hitting the end stops	record SERVO_US_AT_0/180; no buzzing at ends		
T-SWP-1	Powered sweep with the sensor cable fitted	no rubbing, no cable tug at 30° or 150°		
T-THM-1	Temperatures after 30 min running (board regulator, servo)	nothing uncomfortable to touch		
T-LNK-1	Live console over the scanner's own USB cable at 115 200 baud	readings appear, sweep matches the head; note whether opening the port resets the board		
T-LNK-2	Record a session in the console and replay it	replay matches the live view		
T-APP-1	Windows installer + portable build open and connect	app opens; port picker lists the board		
T-APP-2	macOS build opens (after Gatekeeper approval) and connects	as above		
T-FLS-1	Browser flasher (Flash page, Chrome/Edge) installs the release firmware on a CYD	splash screen appears		
T-REL-1	First tagged release carries the source ZIP, manufacturing kit, firmware, desktop apps, SHA256SUMS and attestations	sha256sum -c and gh attestation verify pass		

Sign-off

All gates passed and recorded	☐
Labelled “educational — not a safety device”	☐
Signature / date	